

Using ET

A field guide to ET-based irrigation scheduling in Oregon

— KNOW WHICH ET YOU ARE USING

ET_{ref} · Reference ET

Weather-driven demand from a standard reference surface. ETo uses short grass; ETr uses tall alfalfa.

ET_c · Crop ET

Estimated water use of a specific crop under the conditions represented by the selected crop coefficient.

ET_a · Actual ET

Modeled or measured water consumed under actual field conditions, including management and possible water stress.

— BUILD A TRANSPARENT WATER BALANCE

$$ET_c = K_c \times ET_{ref}$$

Use a crop coefficient matched to the ET reference type and crop stage.

TAW = water capacity × root depth

Use available water capacity and the effective managed root depth.

$$D_2 = D_1 + ET_c - P_e - I_{net}$$

D is depletion; P_e is effective rain. Add other gains or losses when important.

Gross depth = net refill ÷ efficiency

Enter efficiency as a decimal and do not refill beyond root-zone storage.

— FROM NET CROP USE TO A RUN TIME

- $ET_{ref} = 0.24$ in/day and $K_c = 0.85$, so $ET_c = 0.20$ in/day.
- 4-day $ET_c = 0.82$ in; effective rain = 0.05 in.
- Net refill = 0.77 in; at 80% efficiency, **gross depth = 0.96 in.**
- At 0.40 in/hour application, **run time = 2.4 hours.**

— BEFORE USING THE RESULT

- **Confirm crop stage and K_c**
- **Measure applied depth**
- **Check root-zone moisture**
- **Avoid runoff or deep losses**

— ESTIMATE, NOT COMMAND

Station ET may not represent a field with different elevation, wind, advection or crop development. Satellite ET adds spatial context but has model uncertainty. Crop coefficients are generalized curves; soil maps and sensors have scale and placement limits. Use several lines of evidence and correct the schedule when the field disagrees.

— THE 3-CHECK METHOD

- 1 Climate estimates demand**
Reference or actual ET for the question at hand.
- 2 Soil defines storage**
Water capacity, root depth, allowable depletion.
- 3 The field verifies reality**
Moisture, crop condition, applied depth.

COMPANION TOOLS

osuagrohydrology.com
Irrigation calculators and the
Oregon scheduling tool directory

AGHRI LAB

Agrohydrology Research &
Innovation Lab (AGHRI) · Biological &
Ecological Engineering, OSU

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Choose the right data source

Combine sources according to the decision, time scale and level of field detail

— WHAT EACH SOURCE CONTRIBUTES

Weather-station ET

BEST USE Daily crop demand and checkbook scheduling

STRENGTH Frequent local weather updates; established crop curves

WATCH FOR Distance, elevation, advection and crop-stage mismatch

Satellite actual ET

BEST USE Spatial patterns, seasonal use and field comparison

STRENGTH Field-scale coverage and historical context

WATCH FOR Latency, clouds, pixel mixing and model uncertainty

Soil moisture

BEST USE Root-zone status and schedule correction

STRENGTH Direct evidence of storage change at the sensor

WATCH FOR Installation, calibration, placement, sample volume

Field and system checks

BEST USE Confirm crop condition and applied depth

STRENGTH Reveals runoff, nonuniformity, rooting limits, stress

WATCH FOR Needs consistent observation and measurement

— A REPEATABLE WORKFLOW

- 01 Select** Choose a nearby station or field-scale ET product that fits the decision.
- 02 Set** Enter crop stage, K_c , root depth, storage and allowable depletion.
- 03 Track** Update ET, effective rain and measured net irrigation in the balance.
- 04 Verify** Check root-zone moisture, crop condition, depth and uniformity.
- 05 Correct** Reset the balance when field evidence shows the model has drifted.

— COMPANION DIRECTORY

Use the live directory to compare purpose, access requirements, best uses and field cautions for each resource.

osuagrohydrology.com/pages/irrigation-tools.html

- **AgriMet**
Station ET and crop water use
- **WSU Irrigation Scheduler**
Field soil-water balance and timing
- **OpenET**
Satellite actual ET and field reports
- **Web Soil Survey**
Water capacity and soil-map inputs
- **OSU Extension**
Scheduling and soil-moisture guides
- **AGHRI Calculators**
Flow, depth, uniformity and deficit

— PRACTICAL RULE

Do not increase precision in the software beyond the accuracy of the field inputs.

VERIFIED

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Tool interfaces and availability may change over time

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